

Robot data



DROID

real Franka arm episodes; split by recording session



Language-Table

xArm pushing blocks; long unscripted play data

Episode / session

one demonstration; episodes recorded consecutively at one site

End-effector

the tool at the arm's tip, a two-finger gripper

Action block

all commands of one model step, concatenated

the task



3 observed frames
+ future actions →
next 10 frozen DINOv2
patch-feature grids

ShiftWM and evaluation terms



Patch features

frozen DINOv2
16x16 grid, 384-D
each



Transport

each patch takes
content from
observed ones



Transport window

wxw patches in
each of the last S
frames



Gate

per-patch weight:
keep vs. move



Correction

added residual for
new content



Oracle move

observed feature
closest to the
truth



Persistence

copy the last
observed grid



Moving patches

25% with the
largest true
change



Skill

share of
persistence error
removed



Rollout

own predictions
fed back as inputs



Teacher forcing

train on true
past, not own
output



Session split

no recording
session in two
splits

Surgical data



Open-H Hamlyn

dVRK recordings of seven
practice tasks



Practice task

a standard surgeon exercise,
e.g. peg transfer



Phantom

an artificial practice model of
tissue

dVRK

da Vinci Research Kit, surgical robot

Scene camera

external camera, not an endoscope

Simulated suites (plug-in, planning)



PushT

push a T-shaped block to a
target pose



Wall

2-D agent moves through a door
in a wall



TwoRoom

2-D navigation between two rooms
(LeWM)



Reacher

a two-link arm reaches a target
(LeWM)

MPC / CEM

plan, execute the start, re-plan;
sample and refit

World models and baselines

- **V-JEPA 2-AC**
published action-conditioned WM, post-trained on DROID
- **DINO-WM**
published WM on DINOv2 patch features
- ★ **LeWM**
released end-to-end WM; planning reference
- ◆ **Direct**
regresses each future grid:
 $Z_0 + \text{learned residual}$
- ▲ **AR**
residual to the latest grid, rolled out recursively
- **AR-TF**
DINO-WM-style: teacher-forced next step, rolled out